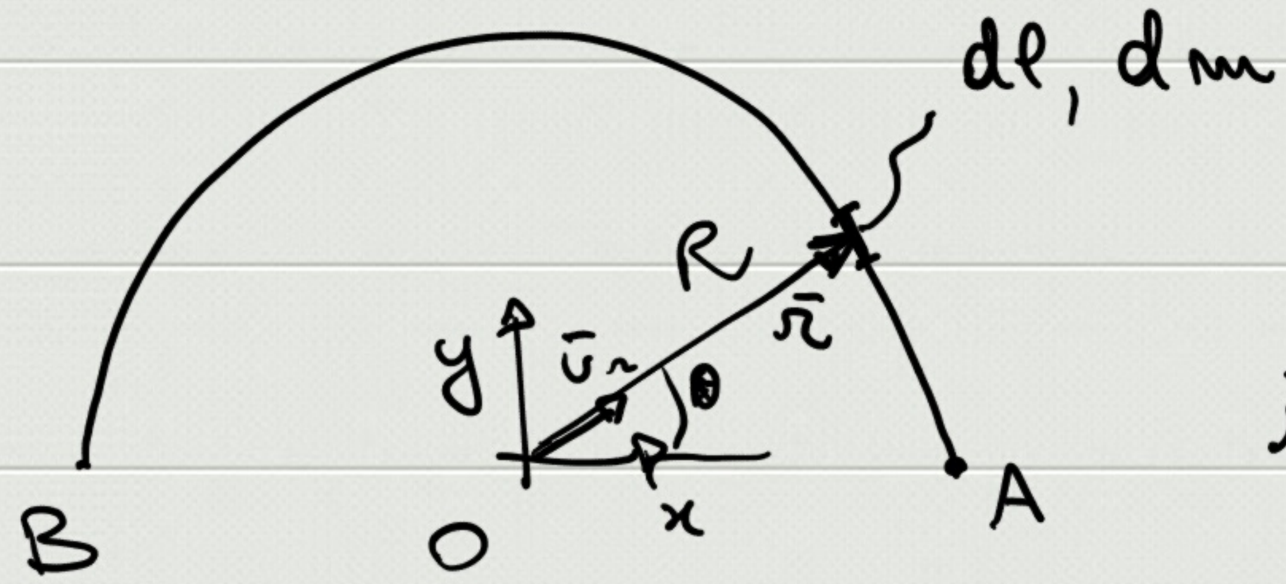


m, R, omogeneo

$\bar{r}_{CM} = ?$

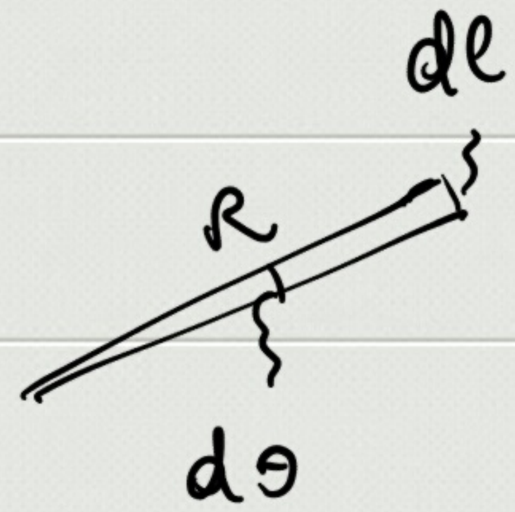
$$\rho_l = \frac{dm}{dl} \qquad \bar{r}_{CM} = \frac{1}{m}\int_{corpo} \bar{r}\, dm$$

$$dm = \rho_l\, dl \qquad \rho_l = cost = \frac{m}{\pi R}$$

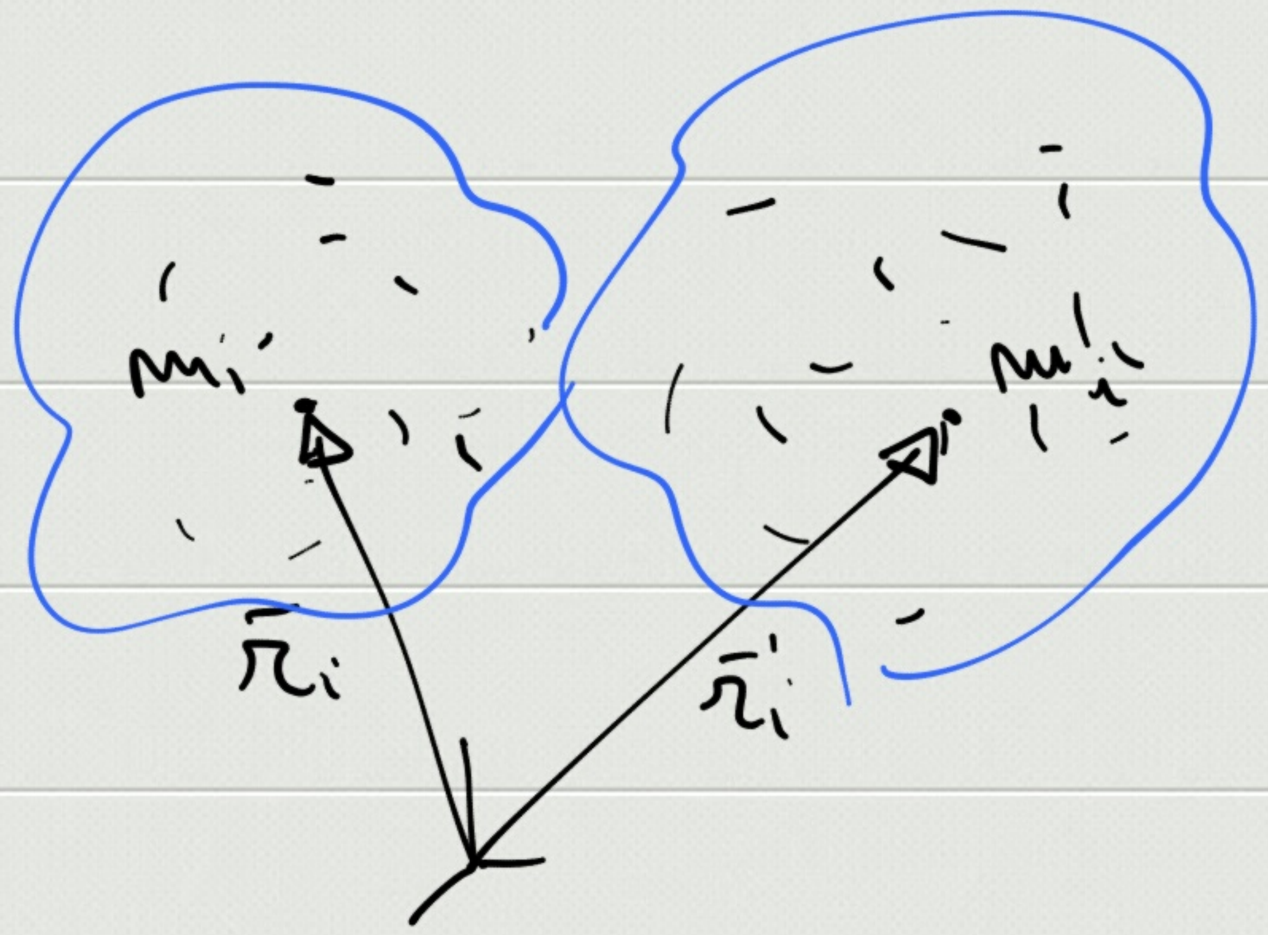
$$\Rightarrow \bar{r}_{CM} = \frac{1}{\cancel{m}}\int_{lungh.} \bar{r}\,\frac{\cancel{m}}{\pi R}\, dl = \frac{1}{\pi \cancel{R}}\int_{lungh} \cancel{R}\,\bar{u}_r\, dl =$$

$$= \frac{1}{\pi}\int_0^{\pi}\left(\cos\theta\,\bar{u}_x + \sin\theta\,\bar{u}_y\right) R\, d\theta =$$

$$= \frac{R}{\pi}\left[\sin\theta\,\bar{u}_x - \cos\theta\,\bar{u}_y\right]_0^{\pi} =$$



$$= \frac{2R}{\pi}\,\bar{u}_y$$



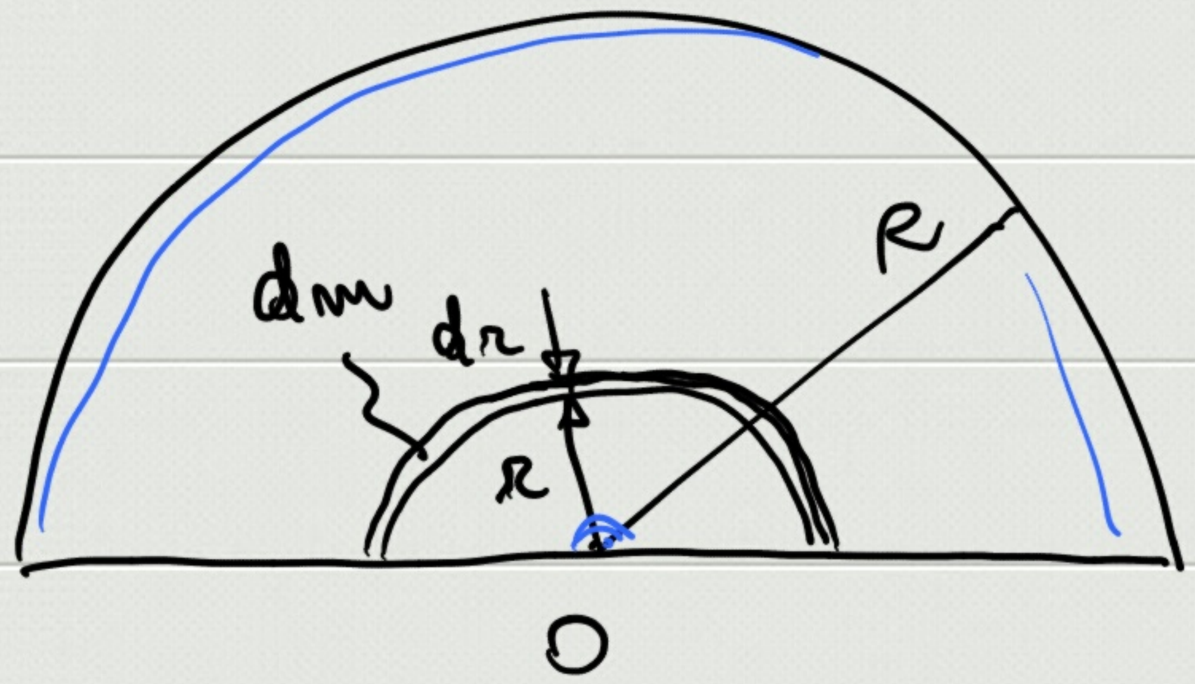
$$\bar{r}_{CM} = \frac{\sum_i m_i \bar{r}_i}{\sum_i m_i} \qquad \bar{r}'_{CM} = \frac{\sum_i m'_i \bar{r}'_i}{\sum_i m'_i}$$

$$\bar{R}_{CM} = \frac{\sum_i (m_i \bar{r}_i + m'_i r'_i)}{\sum_i (m_i + m'_i)} =$$

$$= \frac{\sum_i m_i \bar{r}_i + \sum_i m'_i \bar{r}'_i}{\sum_i m_i + \sum_i m'_i \bar{r}'_i} =$$

$$= \frac{m_{TOT} \bar{r}_{CM} + m'_{TOT} \bar{r}'_{CM}}{m_{TOT} + m'_{TOT}} =$$

$$= \frac{\sum_j m_{TOT,j} \bar{r}_{CM,j}}{\sum_j m_{TOT,j}}$$



Semidisco

R, m, omogeneo

$\bar{r}_{CM} = ?$

$$\bar{r}_{CM,n} = \frac{2r}{\pi}\bar{u}_y$$

$$\bar{r}_{CM,\cancel{a}} = \frac{\int \bar{r}_{CM,n}\, dm_n}{\int dm_n} \qquad (*)$$

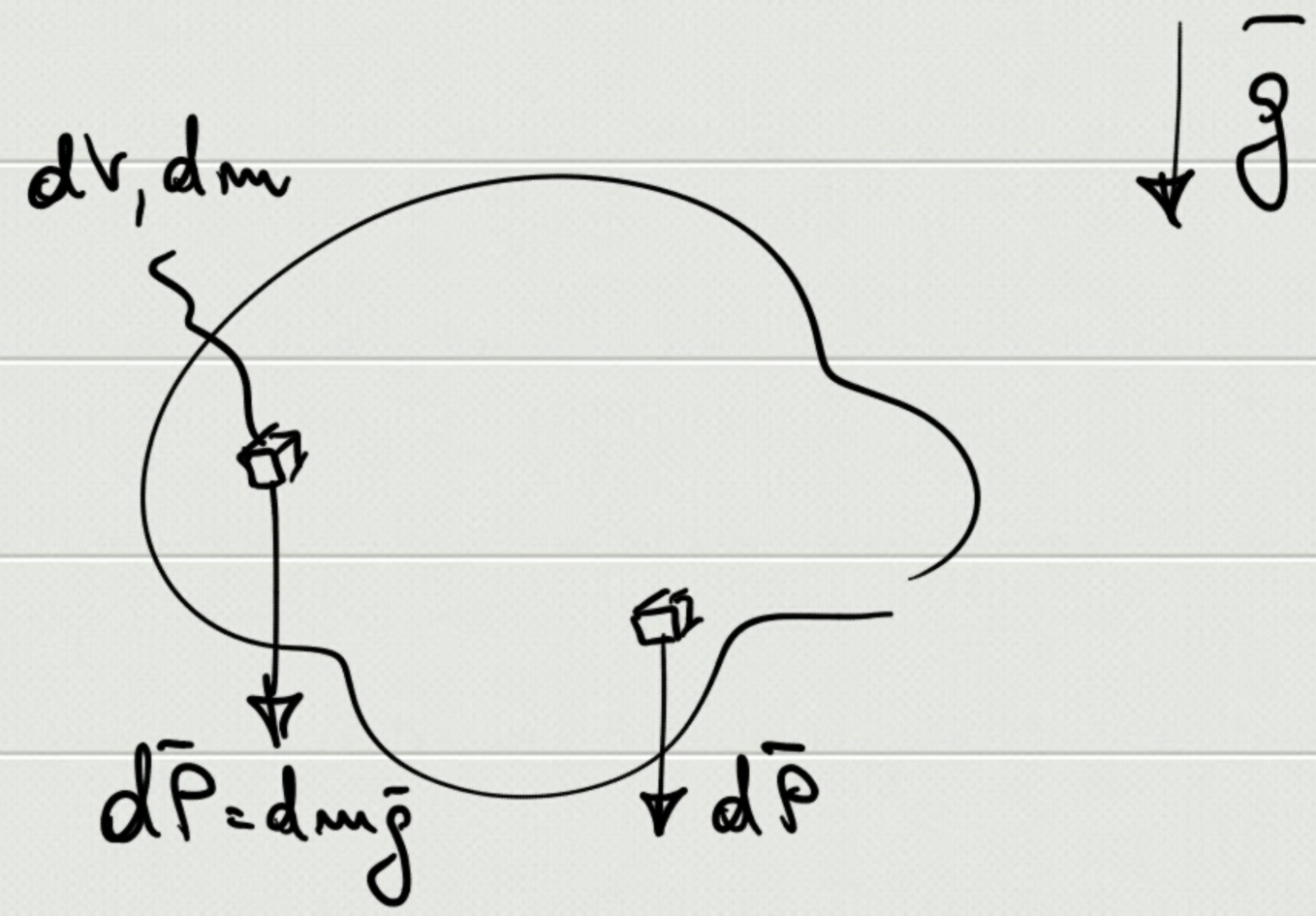
$$\rho_S = \frac{2m}{\pi R^2} \overset{!}{=} \frac{dm}{dS} \Rightarrow dm_n = \rho_S\, dS = \frac{2m}{\pi R^2}\, dS =$$

$$dS = \pi r\, dr \Rightarrow = \frac{2m}{\cancel{\pi} R^2}\cancel{\pi} r\, dr$$

$$(*) \quad \bar{r}_{CM,\cancel{a}} = \frac{1}{\cancel{2m}}\int_0^R \frac{2r}{\pi}\bar{u}_y \cdot \frac{2\cancel{m}}{R^2}\, r\, dr =$$

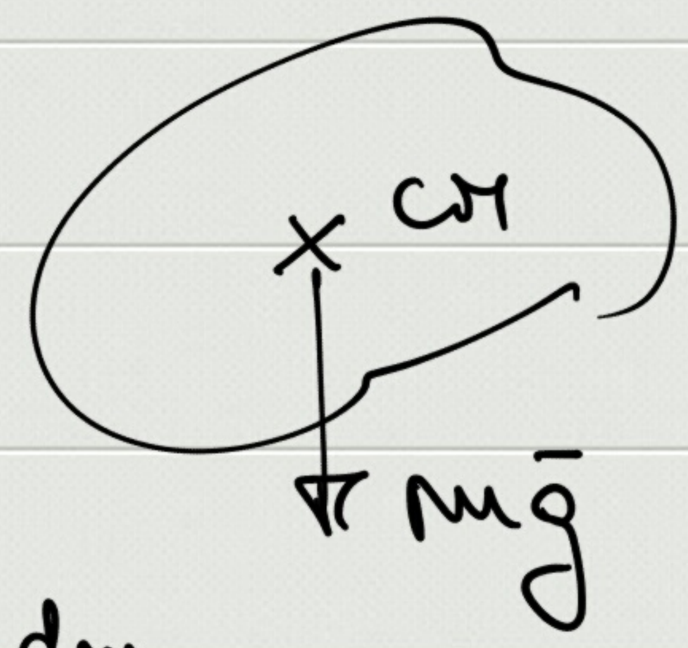
$$= \frac{4}{\pi R^2}\bar{u}_y \int_0^R r^2\, dr = \frac{4}{\pi \cancel{R^2}}\bar{u}_y \cdot \frac{1}{3}R^{\cancel{3}} = \underline{\underline{\frac{4R}{3\pi}\bar{u}_y}}$$

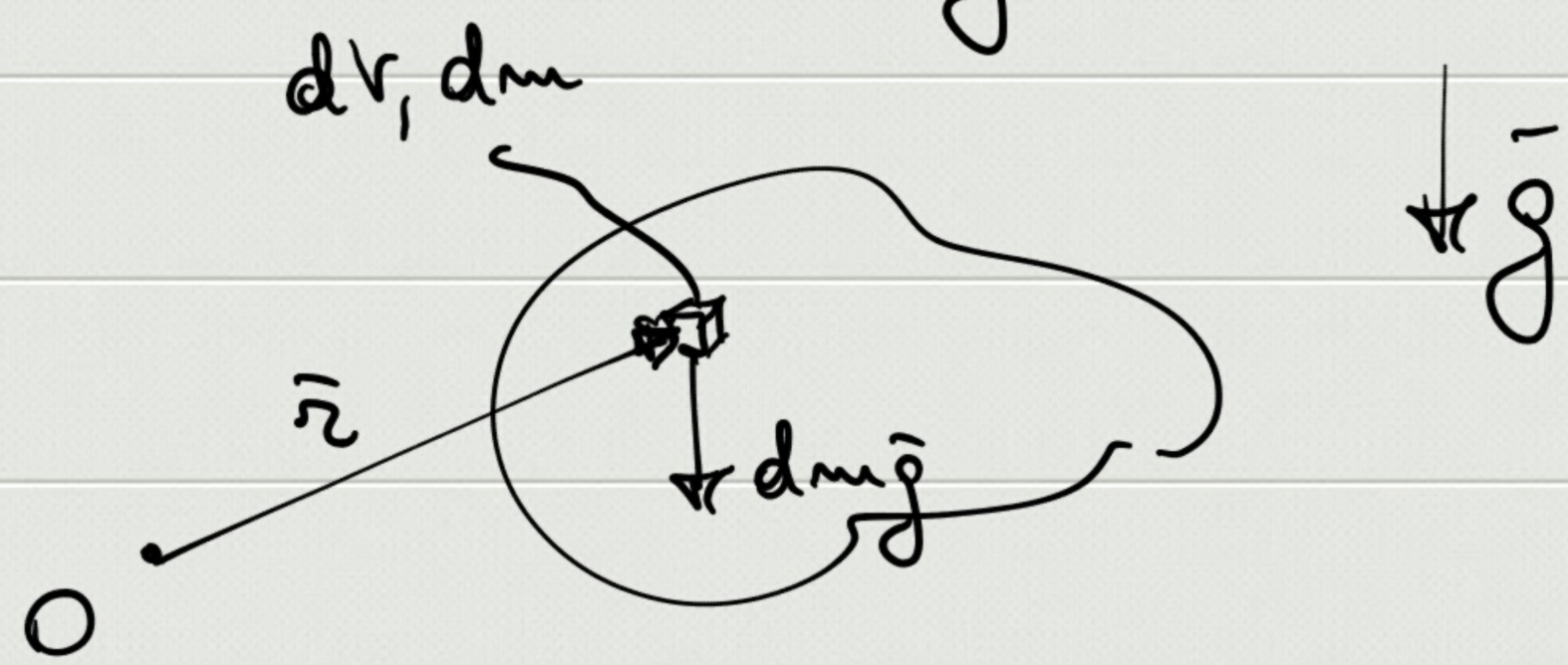
Forza peso



$$\bar{P} = \int_{corpo} d\bar{P} = \int_{corpo} dm\,\bar{g} = \bar{g}\int_{corpo} dm = m\,\bar{g}$$

$$= \bar{R}^{E} = m\,\bar{a}_{CM}$$





$$d\bar{M}_O = \bar{r} \times dm\,\bar{g}$$

$$\bar{M}_O = \int d\bar{M}_O = \int \bar{r} \times dm\,\bar{g} = \left(\int dm\,\bar{r}\right) \times \bar{g} =$$

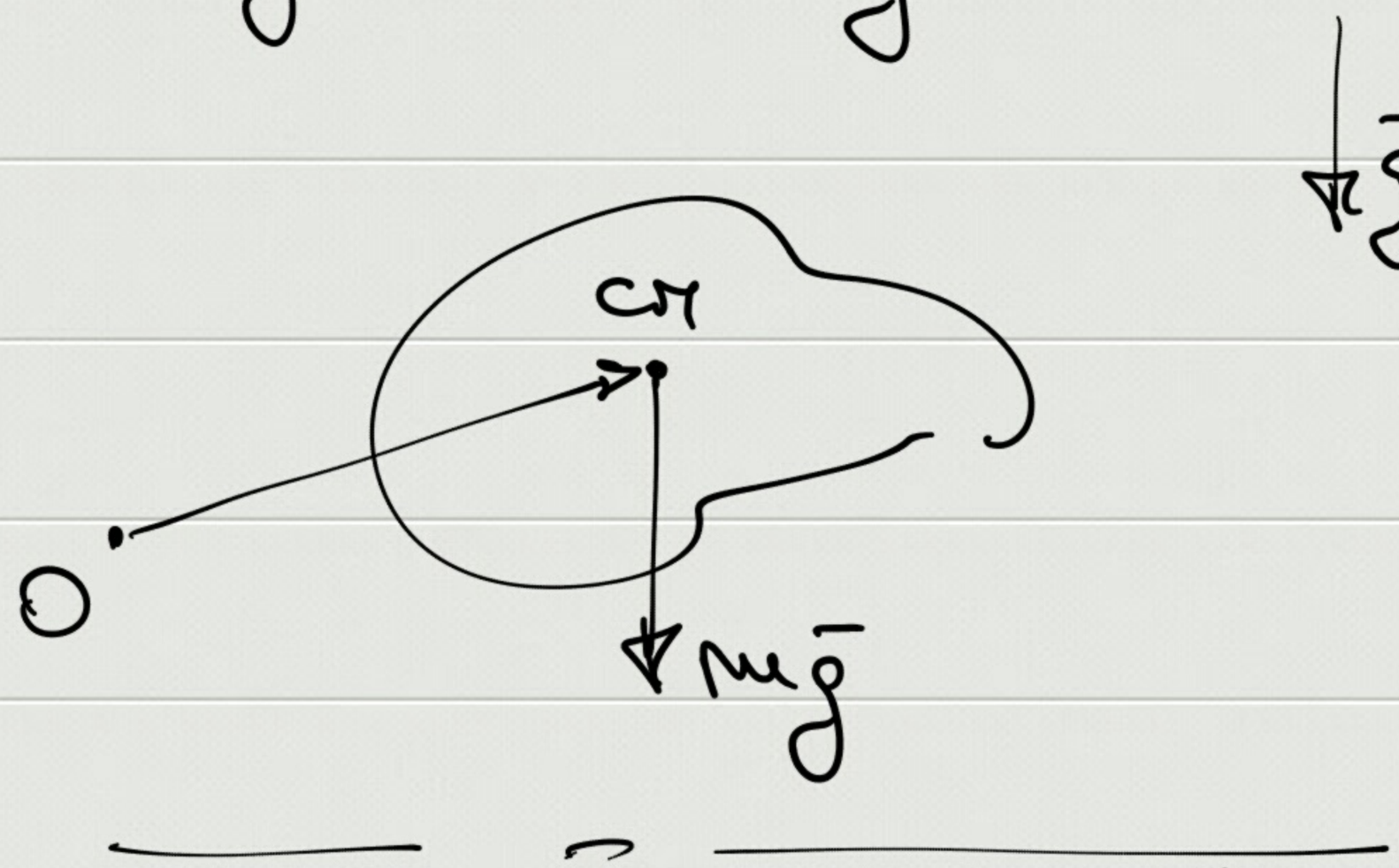
$$\downarrow$$

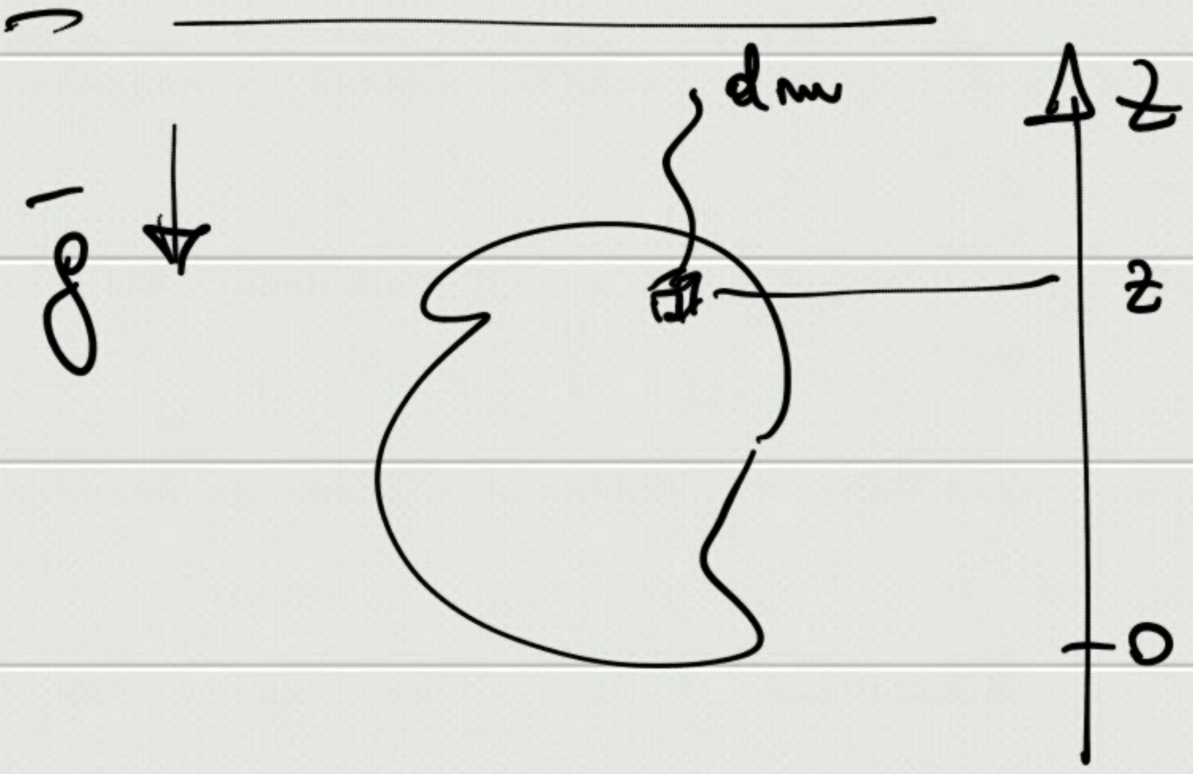
$$m\,\bar{r}_{CM}$$

$$\bar{r}_{CM} = \frac{\int \bar{r}\,dm}{\int dm} = \frac{1}{m}\int \bar{r}\,dm$$

$$= m\,\bar{r}_{CM} \times \bar{g} = \bar{r}_{CM} \times m\bar{g}$$

$\bar{M}_O$





$$E_p = \int dE_p = (*)$$

$$dE_p = dm\, g\, z$$

$$(*) = \int dm\, g\, z = g \int z\, dm = g\, m\, z_{CM}$$

$$z_{CM} = \frac{1}{m} \int z\, dm \quad \Rightarrow$$

$$\Rightarrow \boxed{E_p = m g z_{CM}}$$